

Exploring Unfamiliar Environments: Observe the Environment, Move Freely

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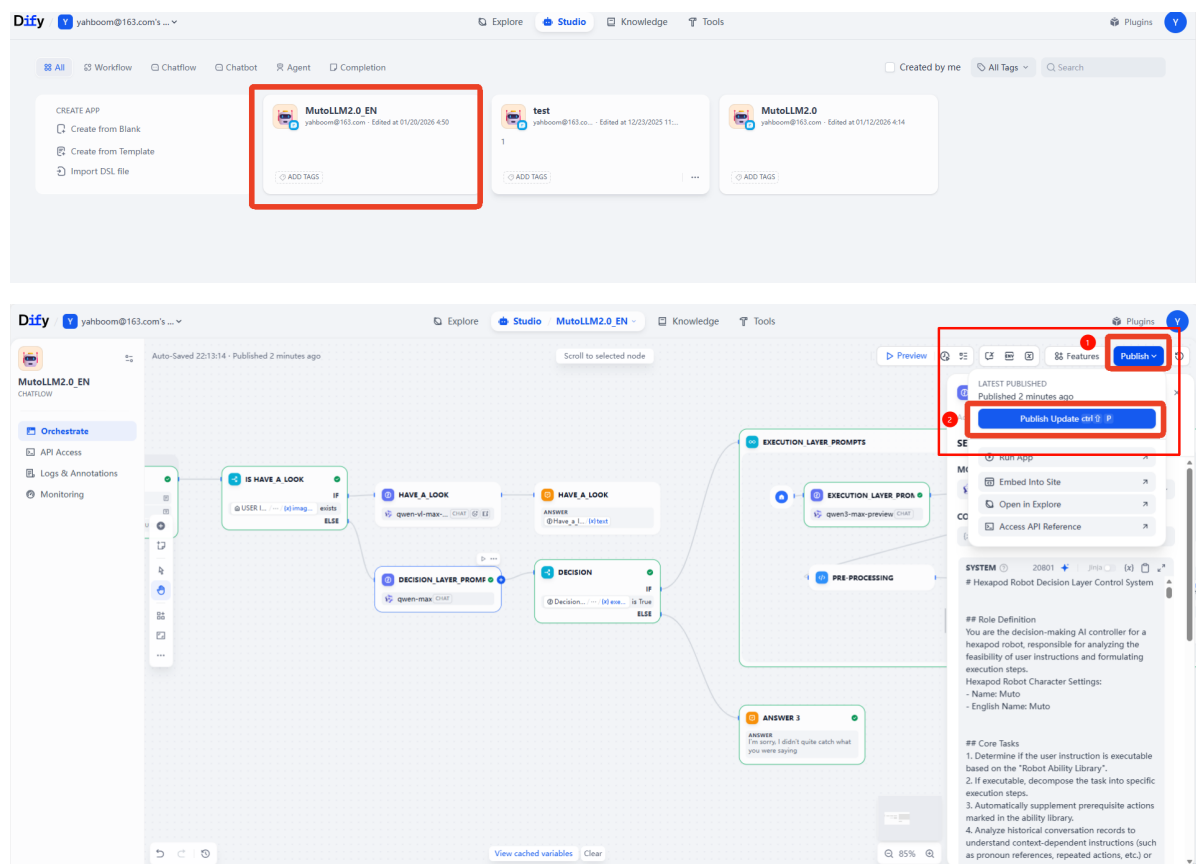
Introduction

In this tutorial, we will demonstrate how to place Muto in an unfamiliar environment and let it explore the environment by randomly combining its existing abilities.

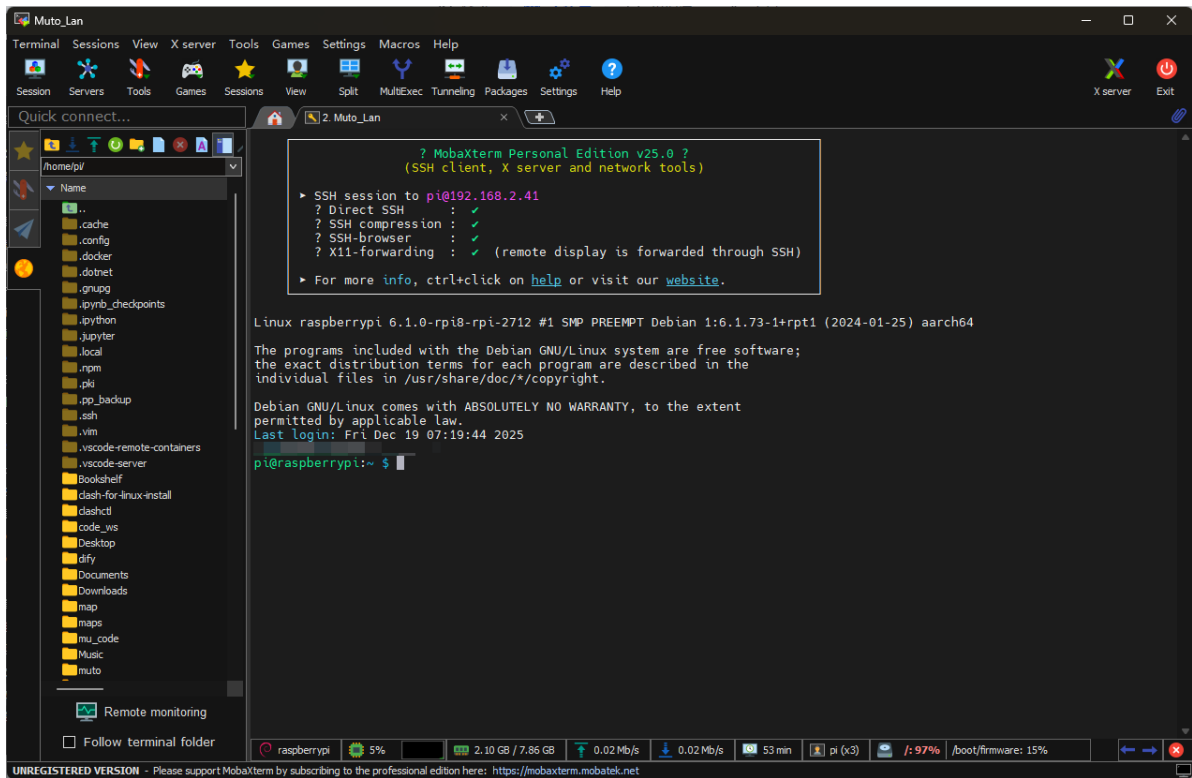
Operation Steps

Preparatory Steps

This lesson requires you to have the Dify terminal open, ensure that Dify is running normally, and that all configuration items have been configured as described in the previous section.



This lesson requires MutoRS to be powered on. Follow the steps outlined in the previous chapters to start the large model terminal within Docker.



```
Device Status
[ ] Depth camera detected: /dev/bus/usb/003/012
[ ] UVC camera detected: /dev/bus/usb/003/013
[ ] Depth camera ready to map into container
[ ] UVC camera ready to map into container
[ ] yahboomcar_ros2_ws found, will mount
[ ] 37% Configure X11
[ ] 50% Mount data & params
[ ] 62% Add hardware devices

Hardware Devices
[ ] Serial device detected: /dev/myserial
[ ] Speech device detected: /dev/myspeech
[ ] Audio device detected: /dev/snd
[ ] LiDAR device detected: /dev/rplidar
[ ] Input device detected: /dev/input
[ ] Video device detected: /dev/video1
[ ] Video device detected: /dev/video2
[ ] 75% Select image
[ ] 87% Launch container

Launch Info
Image: ros2-humble-muto:3.1.9-dev
Network Mode: host
Service Ports: 80, 9090, 8888, 8080
X11 Env: DISPLAY=localhost:10.0

Container started successfully! Container ID: 2027f2d51643
[ ] 100% Launch success

Access Methods
Enter container: docker exec -it 2027f2d51643453252e2b7e516ac12e0adc5d2d21134b276e0080410ba2d76f /bin/bash

Container ready!
-----
ROS_DOMAIN_ID: 38
my_robot_type: Muto | my_lidar: a1
-----
root@raspberrypi:~#
```

```
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239210] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] [INFO]: Started server process [221]
[muto_controller_node-1] [INFO]: Waiting for application startup.
[muto_controller_node-1] [INFO]: Application startup complete.
[muto_controller_node-1] [INFO]: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

Getting Started

The command words we will use in this lesson are as follows:

Hi yahboom, you've arrived in an unfamiliar environment. Please feel free to explore!

Hint Word

To limit Muto and prevent it from moving aimlessly, we used a hint word to indicate a general direction of movement.

The hint word is as follows:

4. **Critical Exploration Tasks**:

- When the user instruction is an open-ended task such as "free exploration" or "wander around," **it is strictly forbidden** to generate fixed templates.
- Please leverage **randomness** and plan a long sequence containing **20-25 steps** to allow the robot to explore autonomously.
- Use the following capabilities in combination:
 - **Visual Perception**: "Initiate intelligent visual analysis to determine what objects are in the scene and analyze the passable direction and distance ahead," "Observe the surrounding environment."
 - **Spatial Perception**: "Use LiDAR to scan distance," "Measure surrounding obstacles," "Determine passability ahead."
 - **Random Movement**:
 - **Do not** hardcode direction and distance (e.g., "Move 1 meter to the left").
 - **Dynamic descriptions** must be used, allowing the execution layer to determine based on real-time data.

- **Recommended Script:** "Based on the visual analysis results or radar data from the previous step, move a safe distance in a random, safe direction."
"Based on environmental data, randomly rotate to an open area."
"While ensuring safety, move randomly and exploratoryly."
- **Core Principle:**
Perception (visual/radar) and action must be performed **alternating**.
- **Visual Analysis Requirements:** Each time a visual analysis is performed, not only must the object be observed, but the analysis must also **clearly** specify the "next possible direction or distance to move".
- **CRITICAL Reporting Requirements:** After each **visual analysis** or **LiDAR detection**, a **step must** be immediately followed by a report of the results.
- **Example Patterns** (For reference only; please generate different combinations each time):
 - Pattern A: Observe (including traffic analysis) -> Announce Result -> Move -> Radar -> Announce Result -> Rotate -> Observe (including traffic analysis) -> Announce Result -> Move -> Action -> Observe
 - Pattern B: Radar -> Announce Result -> Move -> Action -> Observe (including traffic analysis) -> Announce Result -> Move -> Radar -> Announce Result -> Rotate -> Move -> Action